



UNIVERSITÄT **BONN**

Juergen Gall
Shapes
 Computer Vision
 MA-INF 2201 / MA-MOROB-M04
 WS25/26

Organisation



- Exercise sheets (release dates are tentative):
 - Release (40 points): 22.12.
 - Release (40 points): 12.1.
 - Release (20 points): 25.1.
- Overall 280 points + bonus points
 - 140 points are needed to qualify for exam**
- Recap lecture on 23.12.

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Organisation



- 1st Exam: 24.02.2026: 12:30-14:30
 Lecture hall 2, Friedrich-Hirzebruch-Allee 5,
 Hörsaalzentrum
 Registration (HISinOne): **01.12.2025 - 21.12.2025**
- 2nd Exam: 20.03.2026: 9:30-11:30
 Lecture hall 2, Friedrich-Hirzebruch-Allee 5,
 Hörsaalzentrum
 Registration (HISinOne): **01.12.2025 - 13.03.2026**

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Literature



Chapter 17 Models for shape

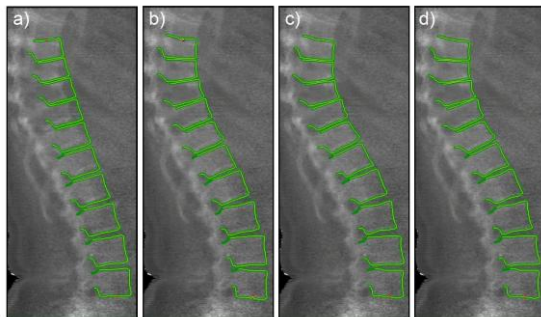
S. Prince. **Computer Vision: Models, Learning, and Inference**. Cambridge University Press 2012

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Motivation: fitting shape model



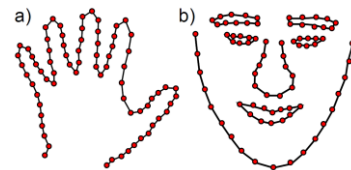
Source: T. Cohen

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Landmark Points



- Landmark points can be thought of as discrete samples from underlying contour
 - Ordered (single continuous contour)
 - Ordered with wrapping (closed contour)
 - More complex organisation (collection of closed and open)

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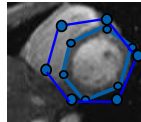
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Recap: Snakes

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- A simple elastic snake is defined by:
 - A set of n points,
 - An internal energy term (tension, bending, plus optional shape prior)
 - An external energy term (gradient-based)
- To use to segment an object:
 - Initialize in the vicinity of the object
 - Modify the points to minimize the total energy



Source: K. Grauman

Probability or not?

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Snakes: minimize cost function

$$E_{total} = E_{internal} + E_{external}$$

Probabilistic view:

$$\begin{aligned} \hat{\mathbf{W}} = \underset{\mathbf{W}}{\operatorname{argmax}} [Pr(\mathbf{W}|\mathbf{x})] &= \underset{\mathbf{W}}{\operatorname{argmax}} [Pr(\mathbf{x}|\mathbf{W})Pr(\mathbf{W})] \\ &= \underset{\mathbf{W}}{\operatorname{argmax}} [\underbrace{\log[Pr(\mathbf{x}|\mathbf{W})]}_{\text{likelihood}} + \underbrace{\log[Pr(\mathbf{W})]}_{\text{prior}}] \end{aligned}$$

Weak priors: smoothness etc...

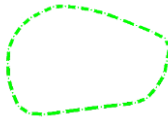
Strong prior: shape

Source: K. Grauman

Snakes

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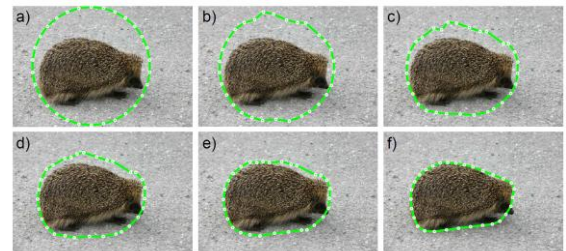
What is the name of the animal?



Source: S. Prince

Snakes

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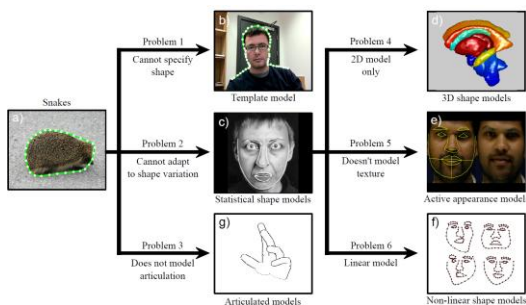


Notice failure at nose – falls between points.
Resulting shape does not represent hedgehog

Source: S. Prince

Relationships between models

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Source: S. Prince

Shape template model

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- Shape based on landmark points $\mathbf{W} = \{\mathbf{w}_n\}_{n=1}^N$
- These points are assumed known
- Mapped into the image by transformation $\mathbf{trans}[\mathbf{w}, \Psi]$
- What is left is to find parameters of transformation Ψ
- Likelihood is based on distance transform:

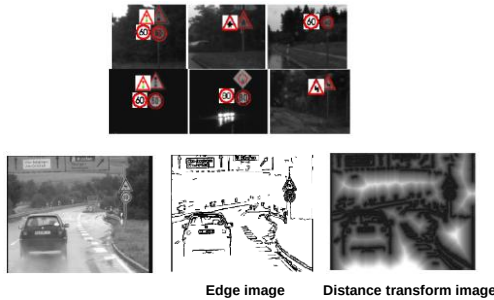
$$Pr(\mathbf{x}|\mathbf{W}, \Psi) \propto \prod_{n=1}^N \exp \left[-(\text{dist}[\mathbf{x}, \mathbf{trans}[\mathbf{w}_n, \Psi]])^2 \right]$$

- No prior on parameters (but could have one)

Source: S. Prince

Recap: Chamfer distance

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Edge image Distance transform image

Source: D. Gavrilu

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Inference

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- Use maximum likelihood approach

$$\hat{\Psi} = \underset{\Psi}{\operatorname{argmax}} L = \underset{\Psi}{\operatorname{argmax}} [\log [Pr(\mathbf{x}|\mathbf{W}, \Psi)]]$$

$$= \underset{\Psi}{\operatorname{argmax}} \left[\sum_{n=1}^N -(\operatorname{dist}[\mathbf{x}, \operatorname{trans}[\mathbf{w}_n, \Psi]])^2 \right]$$

- No closed form solution
- Must use non-linear optimization or brute-force search
- Use chain rule to compute derivatives

$$\frac{\partial L}{\partial \Psi} = - \sum_{n=1}^N \sum_{j=1}^2 \frac{\partial (\operatorname{dist}[\mathbf{x}, \mathbf{w}'_n])^2}{\partial w'_{jn}} \frac{\partial w'_{jn}}{\partial \Psi} \quad \text{where } \mathbf{w}'_n = \operatorname{trans}[\mathbf{w}_n, \Psi]$$

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Source: S. Prince

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Derivative

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Pre-compute Euclidean distance transform D:



Pre-compute gradient by central difference:

$$\frac{\partial}{\partial w'_{jn}} \frac{1}{2} (\operatorname{dist}[\mathbf{x}, \mathbf{w}'_n])^2 \approx D(w'_n) G_j(w'_n) = D(w'_n) \frac{1}{2} (D(w'_n + e_j) - D(w'_n - e_j))$$

Interpolate $\mathbf{w}'$ from nearby pixels

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Linear Transformations

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- Transformations Ψ

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

- Linear 2D transformations can be represented with a 2x2 matrix and are combinations of ...
 - Scale
 - Rotation,
 - Shear
 - Mirror

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Examples

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2D Scaling

$$\mathbf{x}' = \mathbf{s}_x * \mathbf{x}$$

$$\mathbf{y}' = \mathbf{s}_y * \mathbf{y}$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} s_x & 0 \\ 0 & s_y \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

2D Rotate around (0,0)

$$x' = \cos \Theta * x - \sin \Theta * y$$

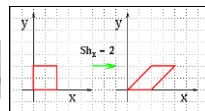
$$y' = \sin \Theta * x + \cos \Theta * y$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos \Theta & -\sin \Theta \\ \sin \Theta & \cos \Theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

2D Shear

$$x' = x + sh_x * y \quad \begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & sh_x \\ sh_y & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$y' = sh_y * x + y$$



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Examples

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2D Mirror about Y axis

$$\begin{aligned} x' &= -x \\ y' &= y \end{aligned}$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

2D Mirror over (0,0)

$$\begin{aligned} x' &= -x \\ y' &= -y \end{aligned}$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

2D Translation

$$\mathbf{x}' = \mathbf{x} + \mathbf{t}_x$$

$$\mathbf{y}' = \mathbf{y} + \mathbf{t}_y$$

NO!

Source: A. Ethier

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Homogeneous coordinates

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To convert to homogeneous coordinates:

$$(x, y) \Rightarrow \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

homogeneous image
coordinates

Converting *from* homogeneous coordinates

$$\begin{bmatrix} x \\ y \\ w \end{bmatrix} \Rightarrow (x/w, y/w)$$

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Source: K. Grauman

Homogeneous Coordinates

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- Q: How can we represent 2d translation as a 3x3 matrix using homogeneous coordinates?

$$\mathbf{x}' = \mathbf{x} + \mathbf{t}_x$$

$$\mathbf{y}' = \mathbf{y} + \mathbf{t}_y$$

- A: Using the rightmost column:

$$\mathbf{Translation} = \begin{bmatrix} 1 & 0 & \mathbf{t}_x \\ 0 & 1 & \mathbf{t}_y \\ 0 & 0 & 1 \end{bmatrix}$$

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Source: A. Efros

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Basic 2D Transformations

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- Basic 2D transformations as 3x3 matrices

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & \mathbf{t}_x \\ 0 & 1 & \mathbf{t}_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Translate

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Scale

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos \Theta & -\sin \Theta & 0 \\ \sin \Theta & \cos \Theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Rotate

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & sh_x & 0 \\ sh_y & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Shear

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Source: A. Efros

2D Affine Transformations

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$$\begin{bmatrix} x' \\ y' \\ w' \end{bmatrix} = \begin{bmatrix} a & b & c \\ d & e & f \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ w \end{bmatrix}$$

- Affine transformations are combinations of
 - Linear transformations, and
 - Translations
- Parallel lines remain parallel



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Source: K. Grauman

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Inference

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- Use maximum likelihood approach

$$\begin{aligned} \hat{\Psi} &= \underset{\Psi}{\operatorname{argmax}} L = \underset{\Psi}{\operatorname{argmax}} [\log [Pr(\mathbf{x}|\mathbf{W}, \Psi)]] \\ &= \underset{\Psi}{\operatorname{argmax}} \left[\sum_{n=1}^N -(\operatorname{dist}[\mathbf{x}, \operatorname{trans}[\mathbf{w}_n, \Psi]])^2 \right] \end{aligned}$$

- No closed form solution
- Must use non-linear optimization or brute-force search
- Use chain rule to compute derivatives

$$\frac{\partial L}{\partial \Psi} = - \sum_{n=1}^N \sum_{j=1}^2 \frac{\partial (\operatorname{dist}[\mathbf{x}, \mathbf{w}'_n])^2}{\partial \mathbf{w}'_{jn}} \frac{\partial \mathbf{w}'_{jn}}{\partial \Psi} \quad \text{where } \mathbf{w}'_n = \operatorname{trans}[\mathbf{w}_n, \Psi]$$

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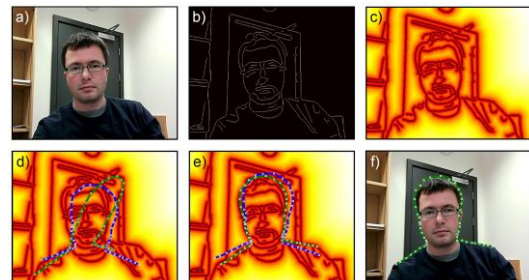
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Source: S. Prince

Shape template model

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Local optimum

Global optimum

Blue initialization, green estimate

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Source: S. Prince

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Iterative closest points



- Maximum likelihood approach

$$\frac{\partial L}{\partial \Psi} = - \sum_{n=1}^N \sum_{j=1}^2 \frac{\partial (\text{dist}[\mathbf{x}, \mathbf{w}'_n])^2}{\partial w'_{jn}} \frac{\partial w'_{jn}}{\partial \Psi} \quad \text{where } \mathbf{w}'_n = \text{trans}[\mathbf{w}_n, \Psi]$$

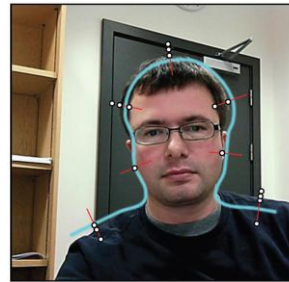
- Iterative closest point:
 - Find closest edge point or line search along normal
 - Estimate Ψ from correspondences (Closed form solution exist)

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Iterative closest points



- Find nearest edge point to each landmark point
- Compute transformation in closed form
- Repeat

Source: S. Prince

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Correspondences



- Line search
- Closest point based on distance transform D:

$$x_n = f(w'_n) = w'_n - \frac{D(w'_n)}{\sqrt{G_x(w'_n)^2 + G_y(w'_n)^2}} (G_x(w'_n)G_y(w'_n))$$

- 1D example:
 - w: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14
 - E: 0 0 0 0 1 0 0 0 0 0 1 0 0 0 0
 - D: 4 3 2 1 0 1 2 3 2 1 0 1 2 3 4
 - G: -1 -1 -1 -1 0 1 1 -1 -1 0 1 1 1 1
 - x: 4 4 4 4 4 4 4 7 10 10 10 10 10 10

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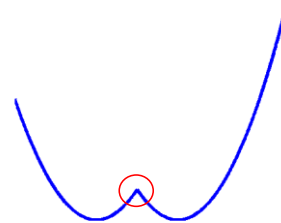
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Derivative



Function is not everywhere differentiable



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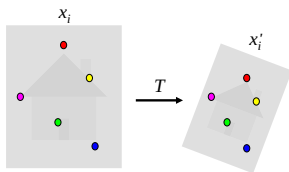
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Alignment problem



Fit the parameters of some **transformation** according to a set of matching feature pairs ("correspondences").



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Source: K. Grauman

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Fitting an affine transformation



$$\begin{bmatrix} x'_i \\ y'_i \end{bmatrix} = \begin{bmatrix} m_1 & m_2 \\ m_3 & m_4 \end{bmatrix} \begin{bmatrix} x_i \\ y_i \end{bmatrix} + \begin{bmatrix} t_1 \\ t_2 \end{bmatrix}$$

$$\begin{bmatrix} m_1 \\ m_2 \\ m_3 \\ m_4 \\ t_1 \\ t_2 \end{bmatrix} = \begin{bmatrix} \vdots \\ \vdots \end{bmatrix}$$

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Source: K. Grauman

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Least squares problem



A $m \times n$ matrix of rank n :

$$Ax = b$$

Solution with pseudo-inverse:

$$A^T(Ax - b) = 0$$

$$x = (A^T A)^{-1} A^T b$$

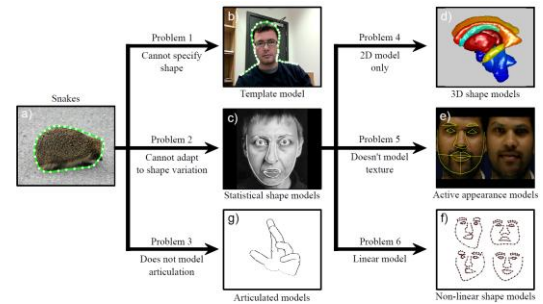
Pseudo-inverse A^+

Solution with SVD:

- SVD: $A = UDV^T$; $A^+ = VD^{-1}U^T$ There are different ways to solve the problem. The optimal method depends on size and sparseness of A
 - $b' = U^T b$
 - $y_i = b'_i / D_{ii}$
 - $x = Vy$
- Specific C++ linear algebra libraries exist!

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Relationships between models



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Statistical shape models



- Also called
 - Point distribution models
 - Active shape models (as they adapt to the image)
- Likelihood:

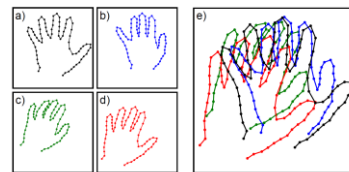
$$Pr(\mathbf{x}_i | \mathbf{w}_i) \propto \prod_{n=1}^N \exp \left[-(\text{dist}[\mathbf{x}_i, \text{trans}[\mathbf{w}_{in}, \Psi_i]])^2 \right]$$

- Prior:

$$Pr(\mathbf{w}_i) = \text{Norm}_{\mathbf{w}_i}[\mu, \Sigma]$$

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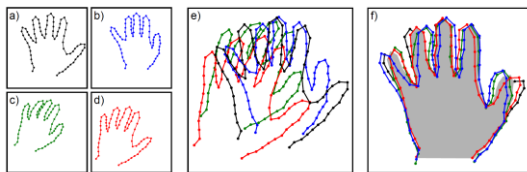
Learning



Training data

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Generalized Procrustes analysis



Training data Before alignment After alignment

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Generalized Procrustes analysis



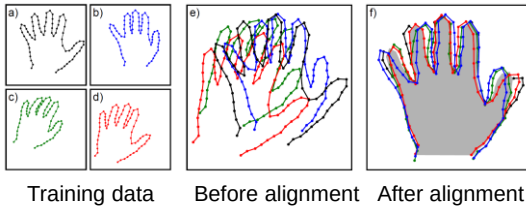
- Given I shapes with N landmarks
- Correspondences between shapes are known (landmarks)
- Take one shape as mean shape
- Iterate:
 - Solve for affine transformation ψ_i for each shape i to the mean $A_i x = b_i$
 - Apply transformation ψ_i to each shape i
 - Compute the mean μ_n of each landmark (mean shape)



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Generalized Procrustes analysis

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Training data

Before alignment

After alignment

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Source: S. Prince

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Inference

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- Map inference:

$$\hat{\mathbf{w}} = \underset{\mathbf{w}}{\operatorname{argmax}} \left[\max_{\Psi} \left[\sum_{n=1}^N -(\operatorname{dist}[\mathbf{x}_i, \operatorname{trans}[\mathbf{w}_n, \Psi]])^2 + \log[\operatorname{Norm}_{\mathbf{w}}[\boldsymbol{\mu}, \boldsymbol{\Sigma}]] \right] \right]$$

- No closed form solution
- Use non-linear optimisation
- Or use ICP approach
- However, many parameters, and not clear they are all needed
- More efficient to use subspace model

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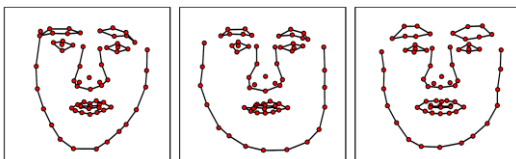
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Source: S. Prince

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Face model

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Three samples from learned model for faces

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Source: S. Prince

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Subspace shape model

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- Generate data from model:

$$\mathbf{w}_i = \boldsymbol{\mu} + \boldsymbol{\Phi} \mathbf{h}_i + \boldsymbol{\epsilon}_i$$

- $\boldsymbol{\mu}$ is the mean shape
- the matrix $\boldsymbol{\Phi} = [\phi_1, \phi_2, \dots, \phi_K]$ contains K basis functions in its columns
- $\boldsymbol{\epsilon}_i$ is normal noise with covariance $\sigma^2 \mathbf{I}$

- Can alternatively write

$$\mathbf{w}_i = \boldsymbol{\mu} + \sum_{k=1}^K \phi_k h_{ik} + \boldsymbol{\epsilon}_i$$

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Source: S. Prince

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Approximating with subspace

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Can approximate a vector $\mathbf{w}$ with a weighted sum of the basis functions

$$\mathbf{w}_i \approx \boldsymbol{\mu} + \sum_{k=1}^K \phi_k h_{ik}$$

Surprising how well this works even with a small number of basis functions

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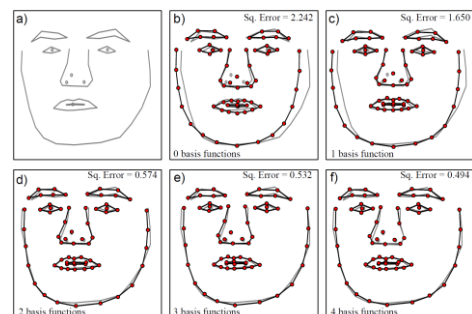
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Source: S. Prince

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Subspace shape model

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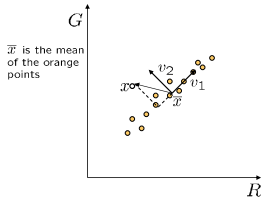
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Source: S. Prince

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Linear subspaces

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- convert x into v_1, v_2 coordinates
 $x \rightarrow ((x - \bar{x}) \cdot v_1, (x - \bar{x}) \cdot v_2)$
- v_2 is distance to line
- v_1 is position along line

- Represent high-dimensional space by linear subspace
- Idea: Estimate new coordinate system from data points

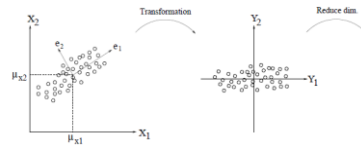
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Dimensionality reduction

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Source: T. Moeslund

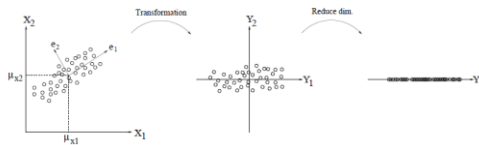
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Dimensionality reduction

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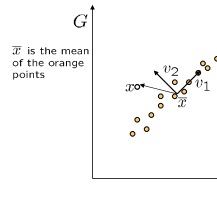
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Linear subspaces

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Consider the variation along direction v among all of the orange points:

$$\text{var}(v) = \sum_{\text{orange point } x} \|(x - \bar{x})^T \cdot v\|^2$$

$$\begin{aligned} \text{var}(v) &= \sum_x \|(x - \bar{x})^T \cdot v\|^2 \\ &= \sum_x v^T (x - \bar{x})(x - \bar{x})^T v \\ &= v^T \left[\sum_x (x - \bar{x})(x - \bar{x})^T \right] v \\ &= v^T A v \quad \text{where } A = \sum_x (x - \bar{x})(x - \bar{x})^T \end{aligned}$$

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Recall: Eigenvalue Problem

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- Maximize objective J (Rayleigh Quotient):

$$J(w) = \frac{w^T S_B w}{w^T S_W w}$$

- Equivalent to

$$\begin{aligned} \min_w \quad & -\frac{1}{2} w^T S_B w \\ \text{s.t.} \quad & w^T S_W w = 1 \end{aligned}$$

- Generalized eigenvalue problem:

$$S_B w = \lambda S_W w$$

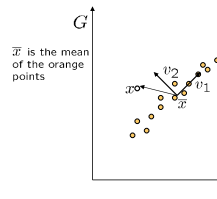
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Linear subspaces

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Consider the variation along direction v among all of the orange points:

$$\text{var}(v) = \sum_{\text{orange point } x} \|(x - \bar{x})^T \cdot v\|^2$$

$$\begin{aligned} \text{var}(v) &= \sum_x \|(x - \bar{x})^T \cdot v\|^2 \\ &= \sum_x v^T (x - \bar{x})(x - \bar{x})^T v \\ &= v^T \left[\sum_x (x - \bar{x})(x - \bar{x})^T \right] v \\ &= v^T A v \quad \text{where } A = \sum_x (x - \bar{x})(x - \bar{x})^T \end{aligned}$$

Solution: v_1 is eigenvector of A with largest eigenvalue
 v_2 is eigenvector of A with smallest eigenvalue

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Principal component analysis (PCA)



- Suppose each data point is N-dimensional
- Same procedure applies:

$$\begin{aligned} \text{var}(\mathbf{v}) &= \sum_{\mathbf{x}} \|(\mathbf{x} - \bar{\mathbf{x}})^T \cdot \mathbf{v}\|^2 \\ &= \mathbf{v}^T \mathbf{A} \mathbf{v} \quad \text{where } \mathbf{A} = \sum_{\mathbf{x}} (\mathbf{x} - \bar{\mathbf{x}})(\mathbf{x} - \bar{\mathbf{x}})^T \end{aligned}$$

- The eigenvectors of $\mathbf{A}$ define a new coordinate system
 - eigenvector with largest eigenvalue captures the most variation among training vectors $\mathbf{x}$
 - eigenvector with smallest eigenvalue has least variation
- We can compress the data by only using the top few eigenvectors
 - corresponds to choosing a "linear subspace"
 - represent points on a line, plane, or "hyper-plane"
 - these eigenvectors are known as the **principal components**

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Principal component analysis (PCA)



- Given data points $\mathbf{w}_i$
- Compute mean μ
- $\mathbf{W} = (\mathbf{w}_1 - \mu, \dots, \mathbf{w}_n - \mu)$
- Eigenvalue decomposition: $\mathbf{W}\mathbf{W}^T = \mathbf{U}\Sigma\mathbf{V}^T\mathbf{V}\Sigma\mathbf{U}^T = \mathbf{U}\mathbf{L}^2\mathbf{U}^T$ ($\mathbf{u}_i$ basis vectors)
- SVD: $\mathbf{W} = \mathbf{U}\Sigma\mathbf{V}^T$

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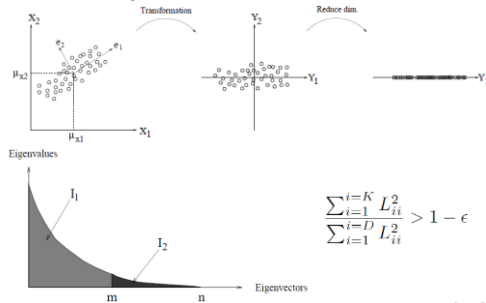
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Principal component analysis (PCA)



- Dimensionality reduction



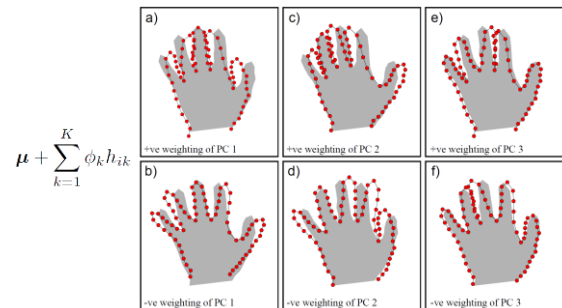
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Source: T. Minka

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Learned hand model



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Source: S. Prince

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Probabilistic PCA



$$\text{Generative eq: } \mathbf{w}_i = \mu + \Phi \mathbf{h}_i + \epsilon_i$$

Probabilistic version:

$$Pr(\mathbf{w}_i | \mathbf{h}_i, \mu, \Phi, \sigma^2) = \text{Norm}_{\mathbf{w}_i}[\mu + \Phi \mathbf{h}_i, \sigma^2 \mathbf{I}]$$

$$\text{Add prior: } Pr(\mathbf{h}_i) = \text{Norm}_{\mathbf{h}_i}[\mathbf{0}, \mathbf{I}]$$

$$\begin{aligned} \text{Density: } Pr(\mathbf{w}_i) &= \int Pr(\mathbf{w}_i | \mathbf{h}_i) Pr(\mathbf{h}_i) d\mathbf{h}_i \\ &= \int \text{Norm}_{\mathbf{w}_i}[\mu + \Phi \mathbf{h}_i, \sigma^2 \mathbf{I}] \text{Norm}_{\mathbf{h}_i}[\mathbf{0}, \mathbf{I}] d\mathbf{h}_i \\ &= \text{Norm}_{\mathbf{w}_i}[\mu, \Phi \Phi^T + \sigma^2 \mathbf{I}] \end{aligned}$$

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Source: S. Prince

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Probabilistic PCA



$$\begin{aligned} &\int \text{Norm}_{\mathbf{w}_i}[\mu + \Phi \mathbf{h}_i, \sigma^2 \mathbf{I}] \text{Norm}_{\mathbf{h}_i}[\mathbf{0}, \mathbf{I}] d\mathbf{h}_i \\ &\quad \text{using } \mathbf{w} = \mu + \Phi \mathbf{h}_i, \quad \Phi^{-1}(\mathbf{w} - \mu) = \mathbf{h}_i \\ &= \int \text{Norm}_{\mathbf{w}_i}[\mu + \Phi \Phi^{-1}(\mathbf{w} - \mu), \sigma^2 \mathbf{I}] \text{Norm}_{\mathbf{w}}[\Phi \mathbf{0} + \mu, \Phi \mathbf{I} \Phi^T] d\mathbf{w} \\ &= \int \text{Norm}_{\mathbf{w}_i}[\mathbf{w}, \sigma^2 \mathbf{I}] \text{Norm}_{\mathbf{w}}[\mu, \Phi \mathbf{I} \Phi^T] d\mathbf{w} \\ &\quad \text{using } \int \text{Norm}_x[a, A] \text{Norm}_x[b, B] dx = \text{Norm}_a[b, A + B] \\ &= \text{Norm}_{\mathbf{w}_i}[\mu, \sigma^2 \mathbf{I} + \Phi \Phi^T] \end{aligned}$$

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Learning PPCA



Learn parameters μ , Φ and σ^2 from data $\{\mathbf{w}_i\}_{i=1}^I$
 where $\mathbf{w}_i = [\mathbf{w}_{i1}^T, \mathbf{w}_{i2}^T, \dots, \mathbf{w}_{iN}^T]^T$.

Learn mean: $\mu = \frac{\sum_{i=1}^I \mathbf{w}_i}{I}$

Then set $\mathbf{W} = [\mathbf{w}_1 - \mu, \mathbf{w}_2 - \mu, \dots, \mathbf{w}_I - \mu]$
 and compute eigen-decomposition

$$\mathbf{W}\mathbf{W}^T = \mathbf{U}\mathbf{L}^2\mathbf{U}^T$$

Choose parameters

$$\hat{\sigma}^2 = \frac{1}{D-K} \sum_{j=K+1}^D L_{jj}^2$$

$$\hat{\Phi} = \mathbf{U}_K (\mathbf{L}_K^2 - \hat{\sigma}^2 \mathbf{I})^{1/2}$$

Source: S. Prince

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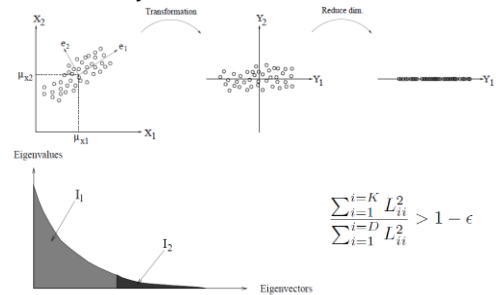
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Principal component analysis (PCA)



• Dimensionality reduction



Source: T. Moeslund

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Properties of basis functions



Learning of parameters based on eigen-decomposition:

$$\mathbf{W}\mathbf{W}^T = \mathbf{U}\mathbf{L}^2\mathbf{U}^T$$

Parameters for representing D-dim. Data by K principal components:

$$\hat{\sigma}^2 = \frac{1}{D-K} \sum_{j=K+1}^D L_{jj}^2 \quad \text{Estimate noise}$$

$$\hat{\Phi} = \mathbf{U}_K (\mathbf{L}_K^2 - \hat{\sigma}^2 \mathbf{I})^{1/2} \quad \text{As PCA but remove noise}$$

Notice that:

- Basis functions in Φ are orthogonal
- Basis functions in Φ are ordered

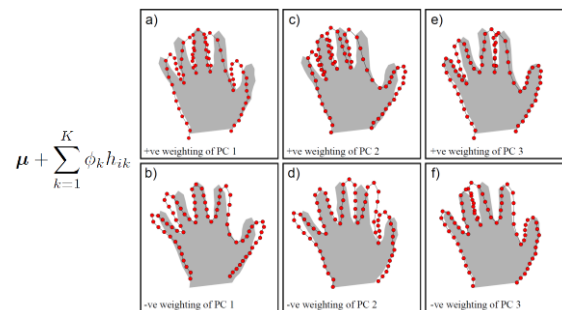
Source: S. Prince

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Learned hand model



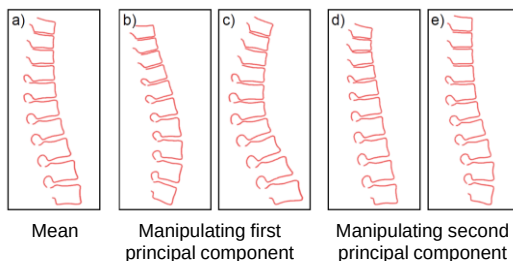
Source: S. Prince

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Learned spine model



Source: T. Cozzini

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Inference



To fit model to an image:

$$\hat{\mathbf{h}} = \underset{\mathbf{h}}{\operatorname{argmax}} \left[\max_{\Psi} \left[\sum_{n=1}^N \underbrace{\left(-\frac{(\operatorname{dist}[\mathbf{x}_i, \operatorname{trans}[\mu_n + \Phi_n \mathbf{h}, \Psi]])^2}{\sigma^2} \right)}_{\text{likelihood}} \right] + \underbrace{\log[\operatorname{Norm}_{\mathbf{h}}[\mathbf{0}, \mathbf{I}]]}_{\text{prior}} \right]$$

ICP Approach:

- Find closest points to current prediction
- Update weightings $\mathbf{h}$
- Find closest points to current prediction
- Update transformation parameters Ψ



Source: S. Prince

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Inference

1. Update weightings $\mathbf{h}$

$$\hat{\mathbf{h}} = \underset{\mathbf{h}}{\operatorname{argmax}} \left[\sum_{n=1}^N \log[Pr(\mathbf{y}_n | \mathbf{h}, \Psi)] + \log[Pr(\mathbf{h})] \right]$$

$$= \underset{\mathbf{h}}{\operatorname{argmax}} \left[\sum_{n=1}^N -(\mathbf{y}_n - \operatorname{trans}[\boldsymbol{\mu}_n + \Phi_n \mathbf{h}, \Psi])^2 / \sigma^2 - \log[\mathbf{h}^T \mathbf{h}] \right]$$

If transformation parameters can be represented as $\mathbf{A}\mathbf{w} + \mathbf{b}$

$$\hat{\mathbf{h}} = \left(\sigma^2 \mathbf{I} + \sum_{n=1}^N \Phi_n^T \mathbf{A}^T \mathbf{A} \Phi_n \right)^{-1} \sum_{n=1}^N \Phi_n^T \mathbf{A}^T (\mathbf{y}_n - \mathbf{A} \boldsymbol{\mu}_n - \mathbf{b})$$

2. Update transformation parameters Ψ

- Using one of the closed form solutions

Source: S. Prince

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Simple version with PCA



1. Initialise the shape parameters, $\mathbf{b}$, to zero (the mean shape).

2. Generate the model point positions using $\mathbf{x} = \bar{\mathbf{x}} + \mathbf{P}\mathbf{b}$

3. Find the pose parameters (X_t, Y_t, s, θ) which best align the model points $\mathbf{x}$ to the current found points $\mathbf{Y}$

$$T_{X_t, Y_t, s, \theta} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} X_t \\ Y_t \end{pmatrix} + \begin{pmatrix} s \cos \theta & -s \sin \theta \\ s \sin \theta & s \cos \theta \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

4. Project $\mathbf{Y}$ into the model co-ordinate frame by inverting the transformation T :

$$\mathbf{y} = T_{X_t, Y_t, s, \theta}^{-1}(\mathbf{Y})$$

5. Project $\mathbf{y}$ into the tangent plane to $\bar{\mathbf{x}}$ by scaling: $\mathbf{y}' = \mathbf{y} / (\mathbf{y} \cdot \bar{\mathbf{x}})$ (optional)

6. Update the model parameters to match to $\mathbf{y}'$

$$\mathbf{b} = \mathbf{P}^T (\mathbf{y}' - \bar{\mathbf{x}})$$

7. If not converged, return to step 2.

T. Cootes. An Introduction to Active Shape Models.

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Fitting model

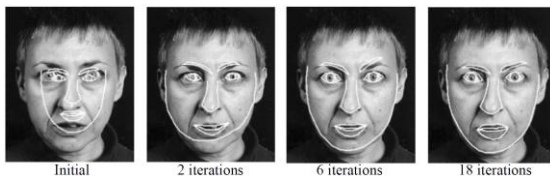


Figure provided by Tim Cootes

Much better to use statistical classifier instead of just distance from edges

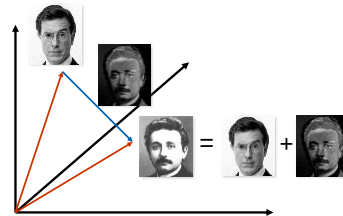
Source: S. Prince

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The space of faces



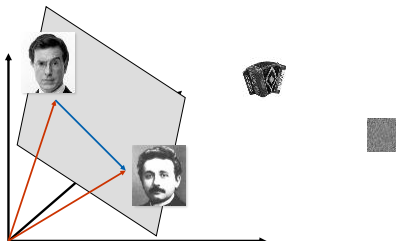
- An image is a point in a high dimensional space
 - An $N \times M$ intensity image is a point in $\mathbb{R}^{NM}$
 - We can define vectors in this space as we did in the 2D case

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Dimensionality reduction



- The set of faces is a "subspace" of the set of images
 - Suppose it is K dimensional
 - We can find the best subspace using PCA
 - This is like fitting a "hyper-plane" to the set of faces
 - spanned by vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_K$
 - any face $\mathbf{x} \approx \bar{\mathbf{x}} + a_1 \mathbf{v}_1 + a_2 \mathbf{v}_2 + \dots + a_K \mathbf{v}_K$

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Eigenfaces

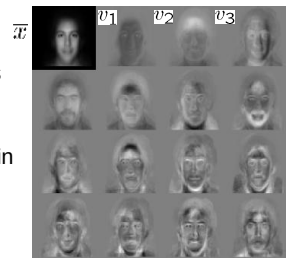


- PCA extracts the eigenvectors of $\mathbf{A}$
 - Gives a set of vectors

$\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots$

- Each one of these vectors is a direction in face space

- what do these look like?



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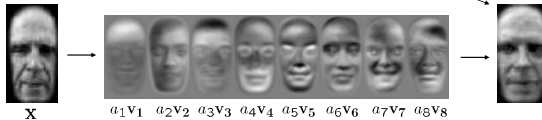
Projecting onto the eigenfaces

- The eigenfaces $\mathbf{v}_1, \dots, \mathbf{v}_K$ span the space of faces
- A face is converted to eigenface coordinates by

$$\mathbf{x} \rightarrow ((\mathbf{x} - \bar{\mathbf{x}}) \cdot \mathbf{v}_1, (\mathbf{x} - \bar{\mathbf{x}}) \cdot \mathbf{v}_2, \dots, (\mathbf{x} - \bar{\mathbf{x}}) \cdot \mathbf{v}_K)$$

$a_1 \quad a_2 \quad a_K$

$$\mathbf{x} \approx \bar{\mathbf{x}} + a_1 \mathbf{v}_1 + a_2 \mathbf{v}_2 + \dots + a_K \mathbf{v}_K$$



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Detection and recognition with eigenfaces

Algorithm

1. Process the image database (set of images with labels)
 - Run PCA—compute eigenfaces
 - Calculate the K coefficients for each image

2. Given a new image (to be recognized) $\mathbf{x}$, calculate K coefficients

$$\mathbf{x} \rightarrow (a_1, a_2, \dots, a_K)$$

3. Detect if $\mathbf{x}$ is a face

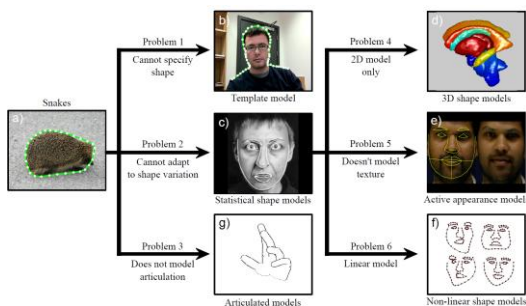
$$\|\mathbf{x} - (\bar{\mathbf{x}} + a_1 \mathbf{v}_1 + a_2 \mathbf{v}_2 + \dots + a_K \mathbf{v}_K)\| < \text{threshold}$$

4. If it is a face, who is it?

- Find closest labeled face in database (nearest-neighbor in K-dimensional space)

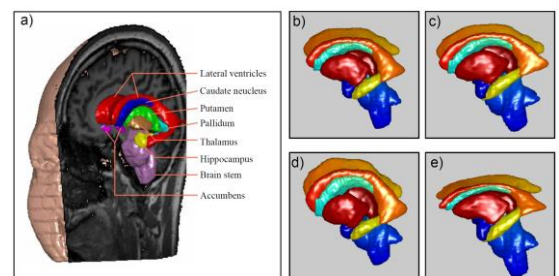
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Relationships between models



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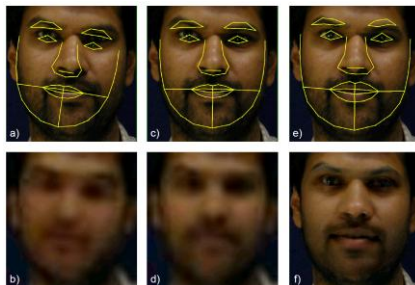
3D shape models



Babalola *et al.* (2008)

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Statistical models for shape and appearance



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Statistical models for shape and appearance

$$\begin{aligned} \Pr(\mathbf{h}_i) &= \text{Norm}_{\mathbf{h}_i}[\mathbf{0}, \mathbf{I}] \\ \Pr(\mathbf{w}_i | \mathbf{h}_i) &= \text{Norm}_{\mathbf{w}_i}[\boldsymbol{\mu}_w + \boldsymbol{\Phi}_w \mathbf{h}_i, \sigma_w^2 \mathbf{I}] \end{aligned}$$

1. We draw a hidden variable $\mathbf{h}$ from a prior
2. We draw landmark points $\mathbf{w}$ from a subspace model

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Statistical models for shape and appearance



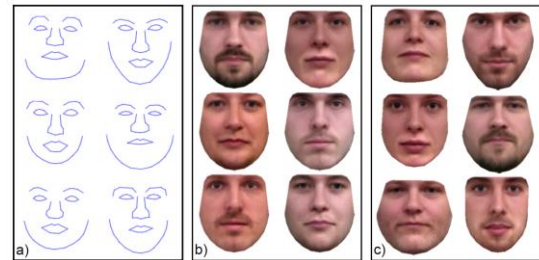
$$\begin{aligned} Pr(\mathbf{h}_i) &= \text{Norm}_{\mathbf{h}_i}[\mathbf{0}, \mathbf{I}] \\ Pr(\mathbf{w}_i | \mathbf{h}_i) &= \text{Norm}_{\mathbf{w}_i}[\boldsymbol{\mu}_w + \boldsymbol{\Phi}_w \mathbf{h}_i, \sigma_w^2 \mathbf{I}] \\ Pr(\mathbf{x}_i | \mathbf{w}_i, \mathbf{h}_i) &= \text{Norm}_{\mathbf{x}_i}[\text{warp}[\boldsymbol{\mu}_x + \boldsymbol{\Phi}_x \mathbf{h}_i, \mathbf{w}_i, \boldsymbol{\Psi}_i], \sigma_x^2 \mathbf{I}] \end{aligned}$$

1. We draw a hidden variable $\mathbf{h}$ from a prior
2. We draw landmark points $\mathbf{w}$ from a subspace model
3. We draw image intensities $\mathbf{x}$.
 - Generate image intensities in standard template shape
 - Transform the landmark points (parameters $\boldsymbol{\Psi}$)
 - Transform the image to landmark points
 - Add noise

Source: S. Prince

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Shape and appearance model



Shape model

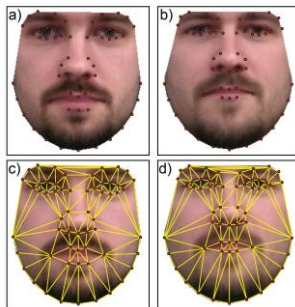
Intensity model

Shape and intensity

Source: M. Stegmann

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Warping images



Piecewise affine transformation

Triangulate image points using Delaunay triangulation.

Image in each triangle is warped by an affine transformation.

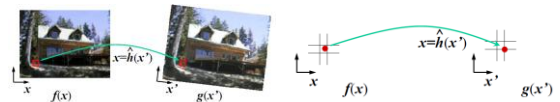
Source: M. Stegmann

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Warping



- GPU/OpenGL rendering
- Inverse map with interpolation:



procedure *inverseWarp*($f, \mathbf{h}, \text{out } g$):

For every pixel x' in $g(x')$

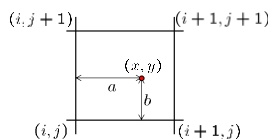
1. Compute the source location $x = \hat{\mathbf{h}}(x')$
2. Resample $f(x)$ at location x and copy to $g(x')$

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Bilinear interpolation



Sampling at $f(x, y)$:



$$\begin{aligned} f(x, y) &= (1-a)(1-b) f[i, j] \\ &\quad + a(1-b) f[i+1, j] \\ &\quad + ab f[i+1, j+1] \\ &\quad + (1-a)b f[i, j+1] \end{aligned}$$

Code from AlphaVision, CMU

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Learning



$$\begin{aligned} Pr(\mathbf{h}_i) &= \text{Norm}_{\mathbf{h}_i}[\mathbf{0}, \mathbf{I}] \\ Pr(\mathbf{w}_i | \mathbf{h}_i) &= \text{Norm}_{\mathbf{w}_i}[\boldsymbol{\mu}_w + \boldsymbol{\Phi}_w \mathbf{h}_i, \sigma_w^2 \mathbf{I}] \\ Pr(\mathbf{x}_i | \mathbf{w}_i, \mathbf{h}_i) &= \text{Norm}_{\mathbf{x}_i}[\text{warp}[\boldsymbol{\mu}_x + \boldsymbol{\Phi}_x \mathbf{h}_i, \mathbf{w}_i, \boldsymbol{\Psi}_i], \sigma_x^2 \mathbf{I}] \end{aligned}$$

Goal is to learn parameters :

$$\{\boldsymbol{\mu}_w, \boldsymbol{\Phi}_w, \sigma_w^2, \boldsymbol{\mu}_x, \boldsymbol{\Phi}_x, \sigma_x^2\}$$

Problem

- We are given the transformed landmark points
- We are given the warped and transformed images

Solution

- Use Procrustes analysis to un-transform landmark points
- Warp observed images to template shape

Source: S. Prince

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Learning

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Now have aligned landmark points $\mathbf{w}_i$ and aligned images $\mathbf{x}_i$, we can learn the simpler model:

$$\begin{aligned} Pr(\mathbf{h}_i) &= \text{Norm}_{\mathbf{h}_i}[\mathbf{0}, \mathbf{I}] \\ Pr(\mathbf{w}_i | \mathbf{h}_i) &= \text{Norm}_{\mathbf{w}_i}[\boldsymbol{\mu}_w + \boldsymbol{\Phi}_w \mathbf{h}_i, \sigma_w^2 \mathbf{I}] \\ Pr(\mathbf{x}_i | \mathbf{h}_i) &= \text{Norm}_{\mathbf{x}_i}[\boldsymbol{\mu}_x + \boldsymbol{\Phi}_x \mathbf{h}_i, \sigma_x^2 \mathbf{I}]. \end{aligned}$$

Can write generative equation as:

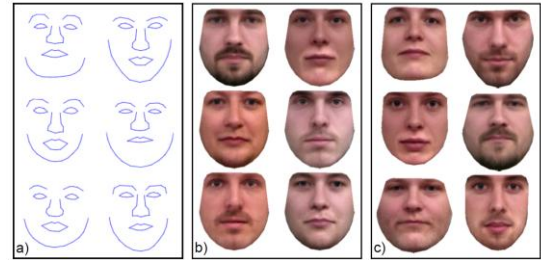
$$\begin{bmatrix} \mathbf{w}_i \\ \mathbf{x}_i \end{bmatrix} = \begin{bmatrix} \boldsymbol{\mu}_w \\ \boldsymbol{\mu}_x \end{bmatrix} + \begin{bmatrix} \boldsymbol{\Phi}_w \\ \boldsymbol{\Phi}_x \end{bmatrix} \mathbf{h}_i + \begin{bmatrix} \boldsymbol{\epsilon}_{wi} \\ \boldsymbol{\epsilon}_{xi} \end{bmatrix}$$

Has the form: $\mathbf{x}' = \boldsymbol{\mu}' + \boldsymbol{\Phi}' \mathbf{h} + \boldsymbol{\epsilon}'$

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Shape and appearance model

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Shape model

Intensity model

Shape and intensity

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Inference

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Likelihood of observed intensities

$$Pr(\mathbf{x} | \mathbf{h}) = \text{Norm}_{\mathbf{x}}[\text{warp}[\boldsymbol{\mu}_x + \boldsymbol{\Phi}_x \mathbf{h}, \boldsymbol{\mu}_w + \boldsymbol{\Phi}_w \mathbf{h}, \boldsymbol{\Psi}], \sigma_x^2 \mathbf{I}]$$

To fit the model use maximum likelihood

$$\begin{aligned} \hat{\mathbf{h}}, \hat{\boldsymbol{\Psi}} &= \underset{\mathbf{h}, \boldsymbol{\Psi}}{\text{argmax}} [\log [Pr(\mathbf{x} | \mathbf{h})]] \\ &= \underset{\mathbf{h}, \boldsymbol{\Psi}}{\text{argmin}} [(x - \text{warp}[\boldsymbol{\mu}_x + \boldsymbol{\Phi}_x \mathbf{h}, \boldsymbol{\mu}_w + \boldsymbol{\Phi}_w \mathbf{h}, \boldsymbol{\Psi}])^T \\ &\quad (x - \text{warp}[\boldsymbol{\mu}_x + \boldsymbol{\Phi}_x \mathbf{h}, \boldsymbol{\mu}_w + \boldsymbol{\Phi}_w \mathbf{h}, \boldsymbol{\Psi}])] \end{aligned}$$

This has the least squares form

$$f[\boldsymbol{\theta}] = \mathbf{z}[\boldsymbol{\theta}]^T \mathbf{z}[\boldsymbol{\theta}]$$

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Inference

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This has the least squares form

$$f[\boldsymbol{\theta}] = \mathbf{z}[\boldsymbol{\theta}]^T \mathbf{z}[\boldsymbol{\theta}]$$

Use Gauss-Newton method or similar

$$\boldsymbol{\theta}^{[t]} = \boldsymbol{\theta}^{[t-1]} + \lambda (\mathbf{J}^T \mathbf{J})^{-1} \frac{\partial f}{\partial \boldsymbol{\theta}}$$

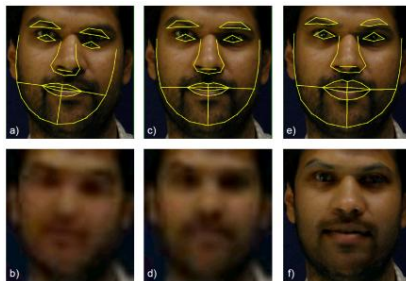
Where the Jacobian $\mathbf{J}$ is a matrix with elements

$$J_{mn} = \frac{\partial z_m}{\partial \theta_n}$$

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Statistical models for shape and appearance

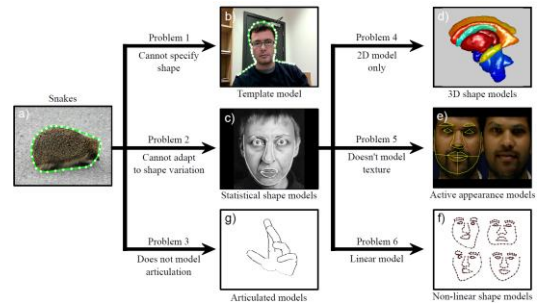
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Relationships between models

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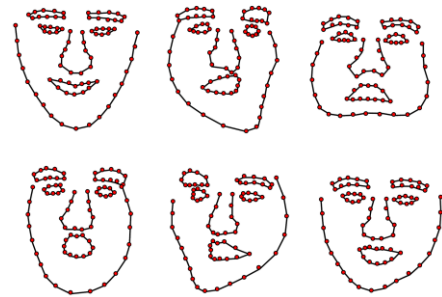
Non-linear models



- The shape and appearance models that we have studied so far are based on the normal distribution
- But more complex shapes might need more complex distributions
 - Could use mixture of PPCAs or similar
 - Or use a non-linear subspace model
- Example: Gaussian process latent variable model (GPLVM) – Computer Vision II

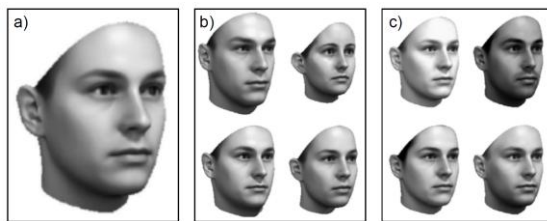
18.12.2025 Jürgen Gall – Institute of Computer Science III – Computer Vision Group Source: S. Prince

GPLVM Shape models



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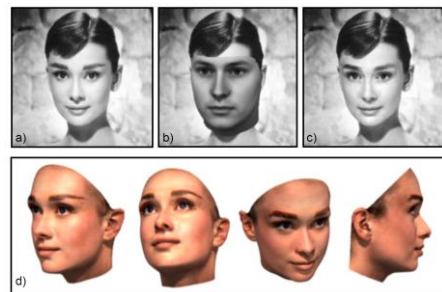
3D morphable models



Adapted from Blanz & Vetter (2003). ©2003 IEEE

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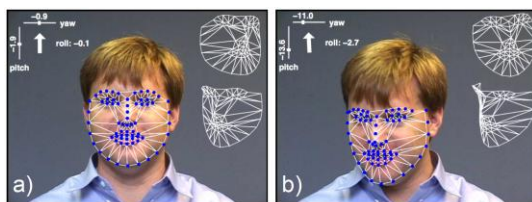
3D morphable models



Adapted from Blanz & Vetter (1999).

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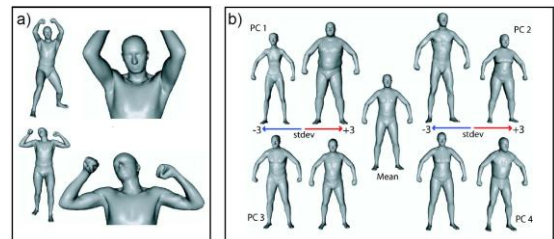
3D morphable models



Matthews *et al.* (2007). ©2007 Springer.

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3D body model

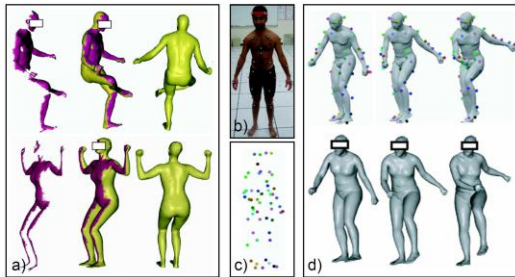


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3D body model applications

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Shape model

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Video editing

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Jain et al. **MovieReshape: Tracking and Reshaping of Humans in Videos.** ToG 2010.

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